

Table 19: **Per-domain mitigation results for Qwen-2.5-7B-Instruct.** **Corr/Incorr:** accuracy (%) on correct/in-correct speaker conditions. DDS computed using dataset-specific C_1 as reference. Deltas relative to C_2 baseline. **Green:** improvement (Avg $\uparrow$, |DDS| $\downarrow$). **Red:** degradation. **Orange:** over-correction from deference to skepticism. $\dagger$ Over-corrected into skepticism (negative DDS). **Bench Avg:** average over 9 standard benchmark datasets. **Total Avg:** average over all 10 datasets including r/AIO. r/AIO exhibits an atypical pattern where SFT and DPO *increase* deference asymmetry (+66.0 and +69.3 DDS respectively), contrasting with their mitigation effect on benchmarks. On benchmarks, SFT achieves the highest accuracy gain ($\uparrow 20.9$) and largest DDS reduction ($\downarrow 20.2$), though it over-corrects on 3 domains (PlausibleQA, HARP, HaluEval). Be Honest prompting achieves comparable DDS reduction ($\downarrow 25.5$) but at the cost of accuracy. DPO provides balanced improvement with strong accuracy gains ($\uparrow 17.1$) and moderate DDS reduction ($\downarrow 6.1$).

Dataset	Condition	Corr	Incorr	Avg	DDS
BBQ	Baseline (C_2)	73.3	55.7	64.5	+40.0
	Be Honest	60.3	75.0	67.7 ($\uparrow 3.2$)	+7.7 ($\downarrow 32.3$)
	Dehumanizing	68.0	62.0	65.0 ($\uparrow 0.5$)	+28.3 ($\downarrow 11.7$)

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Dataset	Condition	Corr	Incrr	Avg	DDS
	SFT	100.0	100.0	100.0 (↑35.5)	+22.3 (↓17.7)
	DPO	99.3	90.3	94.8 (↑30.3)	+31.3 (↓8.7)
TruthfulQA	Baseline (C_2)	83.9	50.6	67.3	+39.7
	Be Honest	72.9	67.0	69.9 (↑2.7)	+12.4 (↓27.3)
	Dehumanizing	79.1	54.6	66.8 (↓0.4)	+31.0 (↓8.7)
	SFT	92.2	94.6	93.4 (↑26.1)	+4.1 (↓35.6)
	DPO	91.8	87.3	89.6 (↑22.3)	+10.9 (↓28.8)
SocialQA	Baseline (C_2)	60.0	80.7	70.3	+39.7
	Be Honest	50.3	88.0	69.2 (↓1.2)	+22.7 (↓17.0)
	Dehumanizing	50.3	86.7	68.5 (↓1.8)	+24.0 (↓15.7)
	SFT	88.3	87.7	88.0 (↑17.7)	+61.0 (↑21.3)
	DPO	83.7	83.7	83.7 (↑13.3)	+60.3 (↑20.6)
AdvisorQA	Baseline (C_2)	63.7	49.0	56.3	+33.0
	Be Honest	56.0	53.3	54.7 (↓1.7)	+21.0 (↓12.0)
	Dehumanizing	56.0	51.0	53.5 (↓2.8)	+23.3 (↓9.7)
	SFT	69.0	55.0	62.0 (↑5.7)	+32.3 (↓0.7)
	DPO	76.3	40.0	58.2 (↑1.8)	+54.7 (↑21.7)
GPQA	Baseline (C_2)	68.7	41.8	55.2	+32.8
	Be Honest	50.0	56.7	53.4 (↓1.9)	−0.7 [†] (↓33.5)
	Dehumanizing	55.2	48.5	51.9 (↓3.4)	+12.7 (↓20.1)
	SFT	64.2	59.0	61.6 (↑6.4)	+11.2 (↓21.6)
	DPO	71.6	46.3	59.0 (↑3.7)	+31.3 (↓1.5)
PlausibleQA	Baseline (C_2)	69.7	34.3	52.0	+28.3
	Be Honest	56.7	46.7	51.7 (↓0.3)	+3.0 (↓25.3)
	Dehumanizing	70.3	34.0	52.2 (↑0.2)	+29.3 (↑1.0)
	SFT	76.0	86.3	81.2 (↑29.2)	−17.3 [†] (↓45.6)
	DPO	85.3	68.3	76.8 (↑24.8)	+10.0 (↓18.3)
HARP	Baseline (C_2)	80.3	27.3	53.8	+25.3
	Be Honest	58.7	49.7	54.2 (↑0.3)	−18.7 [†] (↓44.0)
	Dehumanizing	71.7	35.7	53.7 (↓0.2)	+8.3 (↓17.0)
	SFT	66.0	63.3	64.7 (↑10.9)	−25.0 [†] (↓50.3)
	DPO	79.3	39.3	59.3 (↑5.5)	+12.3 (↓13.0)
HaluEval	Baseline (C_2)	80.0	35.3	57.7	+23.7
	Be Honest	66.3	43.0	54.7 (↓3.0)	+2.3 (↓21.4)
	Dehumanizing	79.0	32.7	55.8 (↓1.8)	+25.3 (↑1.6)
	SFT	98.7	97.0	97.8 (↑40.1)	−19.3 [†] (↓43.0)
	DPO	100.0	96.3	98.2 (↑40.5)	−17.3 [†] (↓41.0)
AMQA	Baseline (C_2)	56.2	74.6	65.4	+6.7
	Be Honest	46.7	82.1	64.4 (↓1.0)	−10.4 [†] (↓17.1)
	Dehumanizing	49.2	78.3	63.7 (↓1.7)	−4.2 (↓10.9)
	SFT	78.8	85.4	82.1 (↑16.7)	+18.3 (↑11.6)
	DPO	75.0	79.6	77.3 (↑11.9)	+20.4 (↑13.7)
r/AIO	Baseline (C_2)	55.0	44.6	49.8	+68.6
	Be Honest	52.1	45.4	48.8 (↓1.1)	+65.0 (↓3.6)
	Dehumanizing	55.0	56.8	55.9 (↑6.1)	+56.4 (↓12.2)
	SFT	88.2	11.8	50.0 (↑0.2)	+134.6 (↑66.0)
	DPO	88.6	8.9	48.8 (↓1.0)	+137.9 (↑69.3)
Bench Avg	Baseline (C_2)	70.6	49.9	60.3	+29.9
	Be Honest	57.5	62.4	60.0 (↓0.3)	+4.4 (↓25.5)
	Dehumanizing	64.3	53.7	59.0 (↓1.3)	+19.8 (↓10.1)
	SFT	81.5	80.9	81.2 (↑20.9)	+9.7 (↓20.2)
	DPO	84.7	70.1	77.4 (↑17.1)	+23.8 (↓6.1)
Total Avg	Baseline (C_2)	67.7	48.9	58.3	+37.3
	Be Honest	56.5	59.2	57.8 (↓0.5)	+15.9 (↓21.4)
	Dehumanizing	62.6	54.3	58.4 (↑0.2)	+26.8 (↓10.5)
	SFT	82.2	74.0	78.1 (↑19.8)	+22.2 (↓15.1)
	DPO	85.1	64.0	74.5 (↑16.3)	+35.2 (↓2.1)

G Extended Cross-Model Analysis

Dataset	N	All 4	None	1–3
AdvisorQA	300	0	178	122
r/AIO	280	43	38	199
AMQA	240	0	156	84
BBQ	300	11	133	156
GPQA	134	0	47	87
HaluEval	300	2	152	146
HARP	300	0	131	169
PlausibleQA	300	0	184	116
SocialIQA	300	2	196	102
TruthfulQA	790	3	428	359
Total	3244	61	1643	1540
%		1.9%	50.6%	47.5%

Table 20: **Cross-model flip consistency.** Number of items that flip (change correctness from C_1 to C_2) in all 4 models, no models, or 1–3 models. Only 1.9% of items flip universally across all models; 49.4% flip in at least one model. r/AIO shows the highest universal flip rate ($43/280 = 15.4\%$), suggesting naturalistic social judgment is particularly susceptible to dialogic effects.

This appendix provides a detailed cross-model analysis.

G.1 Model Summary Statistics

Table 21 provides overall statistics for each model across all domains.

Model	Items	Mean DDS	Flip Rate	DDS Std	Def / Skep
GPT-4o-mini	3,244	+11.2	15.8%	0.568	355 / 156
Qwen-2.5-7B	3,244	+35.0	25.0%	0.679	694 / 117
Gemma-3-12B	3,244	+28.8	20.4%	0.632	589 / 74
GPT-4o	3,244	+2.1	14.8%	0.570	273 / 207

Table 21: Overall model statistics. Def/Skep shows total deference vs. skepticism flips across all domains.

G.2 Flip Rate Comparison by Domain

Table 22 shows the percentage of items that change their judgment between C_1 and C_2 conditions, broken down by deference (accepting incorrect speaker) vs. skepticism (rejecting correct speaker) flips.

G.3 Agreement by Dataset

Table 23 provides item agreement statistics broken down by dataset.

G.4 Consistently Problematic Items

The 61 items that flip in all four models represent “universally problematic” cases. These are distributed as follows: r/AIO (43), BBQ (11), TruthfulQA (3), HaluEval (2), SocialIQA (2). No items

Dataset	Flip Rate (%)			
	GPT-4o-mini	Qwen	Gemma	GPT-4o
AdvisorQA	18.3	19.7	10.3	14.0
r/AIO	35.4	49.6	62.9	52.1
AMQA	13.3	9.2	15.8	5.0
BBQ	20.7	43.3	22.7	9.3
GPQA	12.7	23.9	20.1	34.3
HaluEval	20.0	19.3	17.3	11.7
HARP	6.0	20.3	20.3	26.3
PlausibleQA	10.7	19.7	14.0	6.0
SocialIQA	10.3	15.3	12.3	9.7
TruthfulQA	13.3	25.9	16.6	5.7

Dataset	Deference / Skepticism			
	GPT-4o-mini	Qwen	Gemma	GPT-4o
AdvisorQA	53/2	54/5	30/1	32/10
r/AIO	85/14	117/22	169/7	134/12
AMQA	0/32	13/9	34/4	0/12
BBQ	39/23	101/29	53/15	28/0
GPQA	9/8	27/5	22/5	4/42
HaluEval	37/23	49/9	44/8	28/7
HARP	6/12	50/11	54/7	5/74
PlausibleQA	21/11	52/7	41/1	12/6
SocialIQA	26/5	43/3	27/10	8/21
TruthfulQA	79/26	188/17	115/16	22/23

Table 22: Flip rates and deference/skepticism breakdown by domain and model. r/AIO shows consistently high flip rates across all models (35–63%). GPT-4o exhibits more balanced deference/skepticism ratios than other models.

from AdvisorQA, AMQA, GPQA, HARP, or PlausibleQA flip consistently across all models.

The concentration in r/AIO (70.5% of universal flips) suggests that social judgment tasks are particularly susceptible to dialogic framing regardless of model architecture. These universally problematic items may warrant further investigation to understand what properties make them susceptible to conversational framing.

H Reasoning Failure Taxonomy Results

This appendix details our analysis of reasoning failures underlying 2,465 judgment flips.

H.1 Flip Extraction and Quantitative Summary

Table 24 summarizes judgment flips by model and direction. Deference flips (77.5%) substantially outnumber skepticism flips (22.5%), though GPT-4o shows a more balanced ratio. Table 25 presents aggregate statistics: Qwen-2.5-7B-Instruct exhibits the highest flip rate (25.0%) and strongest deference bias ($DDS = +0.350$), while GPT-4o shows